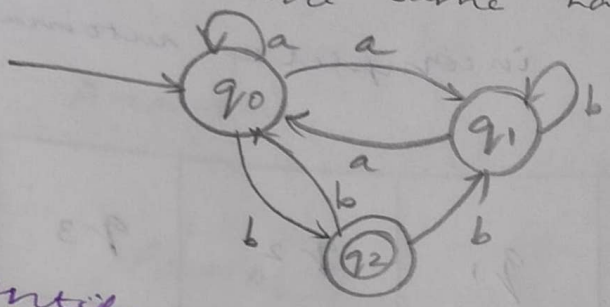
CO1: ASSESSMENT-3

Q) Consider the NFA given below which accepts some language L over the alphabet $\Sigma = \{a, b\}$. construct DFA that represents the same language L.



Identify the NFA:

From the diagram:

* Initial state: q_0
* Final state: q_2
* Alphabet = Σ = a, b

The transitions are:

state	Input a	Input b
q_0	q_0, q_1	q_2
q_1	q_0	q_1, q_2
q_2	–	q_0, q_1

Apply the subset construction Method:

In the subset construction method, each DFA state represents a set of NFA states.

Initial DFA state:

$$A = \{q_0\}$$

1) From $A = q_0$

* on input a: $q_0, q_1 = B$

* On input b: $q_2 = C$.

2) From $B = q_0, q_1$

* On input a: $q_0, q_1 = B$
* On input b: $q_1, q_2 = D$

3) From $C = q_2$

* On input a: $\phi = E$
* On input b: $q_0, q_1, q_2 = B$

4) From $D = q_1, q_2$

* On input a: $q_0 = A$
* On input b: $q_0, q_1, q_2 = F$

5) From $E = \phi$

* On input a: $\phi = E$
* On input b: $\phi = E$

6) From $F = q_0, q_1, q_2$

* On input a: $q_0, q_1 = B$
* On input b: $q_0, q_1, q_2 = F$

DFA Transition Table:

DFA state	NFA state set	Input a	Input b
→ A	q_0	B	C
B	q_0, q_1	B	D
* C	q_2	E	B
D	q_1, q_2	A	F
E	ϕ	E	E
* F	q_0, q_1, q_2	B	F

* Initial state: A
* Final states: C, D, F

A DFA state is final if its set contains the NFA's final state q_2.

Q) Let L be the language of words of length at least 2 whose last two letters are different from each other. For eg: abaab belongs to L, but bb does not.

Understand the language:

The alphabet is

$\Sigma = \{a, b\}$

The conditions are

1) The string must have a length of at least 2.

2) The last two symbols must be different.

Therefore, a string is accepted if it ends with either.

* ab * ba

Eg:

String	Last 2 symbols	Result
ab	ab	Accepted
ba	ba	Accepted
aab	ab	Accepted
bba	ba	Accepted
aa	aa	Rejected
bb	bb	Rejected
abab	ab	Accepted
abba	ba	Accepted.

Construct the NFA:

Use three states:

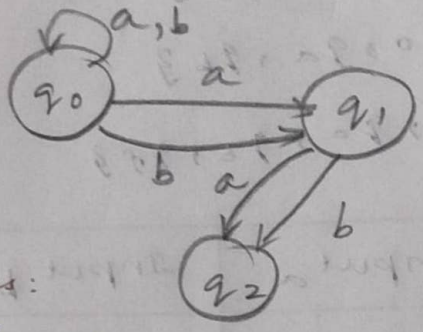
* q_0: Initial states; reads any

number of symbols.

* q_1: guesses that the current symbol is the second-last symbol and reads a.

* q_2: Final state; confirms that the last two symbols are different.

The transitions are.



Transition tables:

Here q_2 is the Final state.

State	Input a	Input b
$\rightarrow q_0$	q_0, q_1	q_0, q_1
q_1	q_2	q_2
$*q_2$	ϕ	ϕ

(b)

It is incorrect for the intended NFA, because q_1 must remember whether it is guessed a or b. Use the corrected NFA below instead.

State	Input a	Input b
$\rightarrow q_0$	q_0, q_a	q_0, q_b
q_a	ϕ	q_f
q_b	q_f	ϕ
$*q_f$	ϕ	ϕ